

Lecture 8 — Bayes Selector: Predicting the Clairvoyant

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8.1 Overview

Lecture 7 converted the fluid LP from a point forecast into a confidence-aware prediction of offline action counts:

$$\text{safe count} = \text{fluid predicted count} - \text{confidence buffer.}$$

We now ask:

If we can predict what hindsight will do, how should we act online?

The main path is

offline-action prediction $\longrightarrow$ compensated coupling $\longrightarrow$ Bayes selection
 $\longrightarrow$ summable disagreements $\longrightarrow O(1)$ regret.

The distinctive idea is not another approximation to a dynamic value function. It is a change in the prediction target:

dynamic programming asks : What is the value of each action?

Bayes Selector asks : What action will hindsight wish I had taken?

8.2 Scope of the guarantee

The strong regret conclusion proved below is conditional. We assume:

1. a known stochastic model of future arrivals;
2. finitely many, sufficiently frequent arrival types;
3. bounded rewards and a known horizon;
4. a stable offline packing LP, in the sense discussed in Lecture 7;
5. exogenous arrivals: the policy does not itself change the future type distribution.

These conditions are substantive. The value of the result is not that constant regret is automatic, but that it identifies a route to very small regret when offline actions can be predicted reliably.

8.3 The offline-action viewpoint

Consider an online allocation state s : remaining time, remaining capacities, and any other information relevant to feasibility. The current request has type j and action set $\mathcal{A}_j$; in network RM this is usually

$$\mathcal{A}_j = \{\text{accept}, \text{reject}\}.$$

For a realized future sample path ω , let

$$\text{OPT}(s, \omega)$$

be the value of an offline optimal allocation beginning from state s . The offline optimum knows all future arrivals in ω .

Definition 1 (Satisfying action). *An action $a \in \mathcal{A}_j$ is satisfying in state s and future realization ω if there exists an offline optimal allocation beginning from (s, ω) that takes action a now.*

The definition allows ties. We do not ask whether every offline optimum takes a ; we ask whether some offline optimum can agree with the online decision without sacrificing hindsight value.

8.4 Compensated coupling

Run the online policy and an offline optimum on the same arrival sequence. At each period:

1. reveal the current type;
2. let the online policy choose its action;
3. if that action is satisfying, select an offline optimum that takes the same action;
4. otherwise, pay the offline benchmark enough compensation to force it to follow the online action.

If the actions agree, no compensation is needed. If they disagree in a consequential way, the compensation measures the lost offline value caused by forcing agreement.

Let C_t be the compensation at time t . Telescoping the coupled benchmark values gives

$$\text{Reg}(T) = \mathbb{E} \left[\sum_{t=1}^T C_t \right]. \quad (8.1)$$

For bounded-reward packing problems, there is a constant $c_{\max}$ such that

$$C_t \leq c_{\max} \mathbf{1}\{a_t \text{ is not satisfying}\}.$$

Therefore

$$\text{Reg}(T) \leq c_{\max} \sum_{t=1}^T \mathbb{P}(a_t \text{ is not satisfying}). \quad (8.2)$$

This is the reason to predict offline actions:

$$\text{control costly disagreement with hindsight} \implies \text{control regret}.$$

8.5 Bayes selection

Condition on the current state s and type j . For action $a \in \mathcal{A}_j$, define its conditional disagreement probability

$$q_t(s, j, a) = \mathbb{P}(a \text{ is not satisfying for offline} \mid s, J_t = j). \quad (8.3)$$

The ideal Bayes Selector chooses

$$a_t \in \arg \min_{a \in \mathcal{A}_j} q_t(s, j, a). \quad (8.4)$$

The word “Bayes” refers to averaging over the stochastic future conditional on current information. Unlike UCB, the selector is not learning an unknown reward distribution. It is predicting an offline action under a probabilistic model of future arrivals.

8.6 The two-value example revisited

Recall the one-resource setting from Lecture 7. There are τ future periods, remaining capacity b , high requests worth 2, and low requests worth 1. Let

$$H_\tau \sim \text{Binomial}(\tau, 1/2)$$

count future high requests.

For a current low request:

- if we accept, offline disagrees when $H_\tau \geq b$, because the low request displaces a future high;

- if we reject, offline disagrees when $H_\tau \leq b - 1$, because there was enough capacity for the low request after all.

Thus

$$q_{\text{accept}}(\tau, b) = \mathbb{P}(H_\tau \geq b), \quad (8.5)$$

$$q_{\text{reject}}(\tau, b) = \mathbb{P}(H_\tau \leq b - 1). \quad (8.6)$$

The ideal Bayes Selector chooses the smaller tail probability.

Notice what it does not compute: it does not recursively calculate $V_\tau(b)$ at every state. It asks a classification question about hindsight:

On which side of the current action will the clairvoyant usually lie?

8.7 Implementing the prediction with safe budgets

Exact disagreement probabilities such as (8.5)–(8.6) may be difficult in a network. Lecture 7 gave a tractable surrogate.

With τ periods remaining, solve the fluid LP and obtain predicted offline action counts

$$x_{ja}^{\text{fl}}(\tau).$$

Choose a confidence buffer $\kappa u(\tau)$ and define

$$L_{ja}(\tau) = [x_{ja}^{\text{fl}}(\tau) - \kappa u(\tau)]^+. \quad (8.7)$$

If $L_{ja}(\tau) > 0$, then on the confidence event some selected offline optimum still wants action a on type j . A confidence-buffered selector chooses an action with positive safe budget and decrements that budget after using it.

The policy is therefore simple to state:

Take actions that remain inside a confidence-certified offline budget.

8.8 The clean-event proof

Let $Z(\tau)$ be the future arrival-count vector and define

$$\mathcal{E}_\tau = \{\|Z(\tau) - \tau\lambda\|_\infty \leq u(\tau)\}. \quad (8.8)$$

Choose $u(\tau)$ so that, for example,

$$\mathbb{P}(\mathcal{E}_\tau^c) \leq \frac{C_0}{\tau^2}. \quad (8.9)$$

The proof has four steps.

8.8.1 Step 1: condition on the clean event

On $\mathcal{E}_\tau$, the future type counts lie in the confidence region used to construct the safe budgets.

8.8.2 Step 2: reason deterministically

LP stability implies

$$x_{ja}^{\text{off}} \geq x_{ja}^{\text{fl}} - \kappa u(\tau) = L_{ja}(\tau)$$

for a suitable selected offline optimum. Therefore an action taken from a positive safe budget is satisfying on the clean event, except for bookkeeping and boundary effects near the end of the horizon.

This is the deterministic heart of the argument:

$$\boxed{\text{positive safe budget} + \text{clean event} \implies \text{agreement with some offline optimum.}} \quad (8.10)$$

8.8.3 Step 3: charge confidence failures

Away from the endgame, a consequential disagreement can occur only if $\mathcal{E}_\tau$ fails. Its probability is therefore bounded by C_0/τ^2 .

8.8.4 Step 4: control the endgame

When only a bounded number of periods or units remain, confidence buffers may be larger than the predicted action counts. We stop using the asymptotic argument and bound the remaining loss directly by a problem-dependent constant.

8.9 Why summability matters

Suppose the probability of a consequential disagreement with τ periods remaining satisfies

$$q_\tau \leq \frac{C_0}{\tau^2}.$$

Then

$$\sum_{\tau=1}^T q_\tau \leq C_0 \sum_{\tau=1}^{\infty} \frac{1}{\tau^2} < \infty. \quad (8.11)$$

Combining (8.2), (8.10), (8.11), and the bounded endgame yields the qualitative guarantee.

Theorem 8.1 (Informal conditional constant-regret guarantee). *Under the assumptions in Section 8.2, a confidence-buffered Bayes-selector policy has*

$$\text{Reg}(T) \leq C,$$

where C depends on the instance but not on T .

The result is strong because the benchmark is the offline clairvoyant. It is conditional because accurate, stable prediction of offline actions is doing real work.

Returning to the three scales from Lecture 7, a fluid policy can capture order- T value while losing order $\sqrt{T}$. Under the stated conditions, Bayes selection uses concentration and stable offline predictions to target the order-1 scale:

Bayes selection uses scale to go beyond first-order optimality.

8.10 UCB and Bayes Selector

The lectures now share a common proof architecture:

	UCB	Bayes Selector
uncertain object	arm mean μ_i	future arrivals Z
confidence object	interval around μ_i	region around Z
decision rule	optimistic arm	offline-predicted action
key implication	$\Delta_i \leq 2r_i$	disagreement implies confidence failure
regret control	few bad pulls	few costly disagreements

In UCB, a suboptimal arm can be selected only while uncertainty is large. In Bayes Selector, an online action can disagree with hindsight only when confidence fails, the optimization is unstable, or the system is in a bounded endgame.

The final equation to remember is

probabilistic clean event $\implies$ deterministic control of bad actions $\implies$ regret bound.

8.11 What can break the conclusion?

Constant regret should not be expected when important types become too rare, the horizon is unknown and cannot be handled robustly, the offline solution is unstable, rewards are unbounded, or the policy changes the future arrival distribution.

The final failure is the bridge to Lecture 9. In revenue management with substitution, availability changes customer choices. Those sales may then be used to re-estimate demand. The arrival model is no longer cleanly exogenous, and confidence around the wrong model cannot repair the problem.

8.12 Takeaways

1. Compensated coupling converts regret into the cost of disagreements with a dynamic offline benchmark.
2. Bayes Selector chooses actions with small conditional disagreement probability.
3. Safe budgets implement offline-action prediction using fluid counts and confidence buffers.
4. On the clean event, positive safe budgets certify satisfying actions.
5. Summable confidence failures plus a bounded endgame can yield instance-dependent $O(1)$ regret, going beyond first-order fluid optimality.
6. The same clean-event proof architecture appeared in UCB, but the uncertain object and chosen action are different.

Suggested reading

Vera and Banerjee, *The Bayesian Prophet: A Low-Regret Framework for Online Decision Making*, especially Sections 3–4, for compensated coupling, disagreement probabilities, and Bayes selection. Banerjee and Freund, *Good Prophets Know When the End Is Near*, especially Sections 2–3, gives the direct confidence-bound formulation used here and explains why arrival thickness, horizon information, and LP stability matter for uniform additive loss.